

INTERPRETABLE ANOMALY DETECTION

How explainability helps detect fraudulent transactions or cyberattacks.

Julian Dinnissen

s1135596

Table of Contents

Introduction & Problem definition -----	2
Research gap -----	2
Methods -----	3
Preprocessing & Random Forest-----	3
SHAP -----	3
iBreakDown -----	3
Model Comparison -----	4
Evaluating Methods -----	4
Model Fidelity-----	4
Cross-Model Consistency-----	4
Results -----	5
Discussion -----	6
References -----	7

Introduction & Problem definition

Anomalies can be found in every dataset and are defined as identifiable elements different from normal data points. In the field of data analysis, anomaly detection (AD) is the process of finding such anomalies or outlying patterns in a data set. This process is essential in a range of areas such as cybersecurity, health care, and finance.

In cybersecurity, anomaly detection is used for finding unusual network traffic data mainly detecting cyberattacks. Preventing attacks is traditionally done by a rule-based approach. In recent years attack methods have significantly improved to a level where simple detection methods no longer suffice. Current AD methods use more advanced artificial intelligent (AI) methods in the field of machine learning and specifically deep learning (DL) (Gómez et al., 2023).

While modern machine learning models like Neural Networks or Random Forests have a high accuracy in class prediction, they are viewed as “black boxes”. This means that the underlying operations that result in the model’s output cannot be traced to a single decision. In environments like cyber-security, the lack of transparency prevents analysts from identifying specific network features that are part of a specific threat.

This paper discusses the methods and results of applying XAI methods to existing AI models, with the research question: **To what extent do SHAP and iBreakDown (dalex) provide interpretable explanations for an anomaly detection Random Forest model, and which model serves the use case of detecting network intrusions the best?**

Research gap

While recent research has underscored the importance of XAI in the cybersecurity sector (James, 2024) and shows increased performance in recall and precision in an industrial setting (Gómez et al., 2023), a significant gap remains regarding algorithmic consistency and cross-model faithfulness of explanations. Most studies in the field focus on one XAI framework (usually SHAP or LIME) (Almheiri et al., 2025), without exploring different mathematical approaches, such as the difference between Additive vs. Sequential attribution. This lack of model comparison is an interesting gap in the field, when explored, would allow insight into the security contribution of each model’s explanation. Furthermore, there is a shortage of validation whether these models truly represent the model’s internal operations or if they merely provide human-viewable visualizations, which could lead to misplaced trust. This project addresses this gap by comparing the analysis of SHAP and iBreakDown (dalex) explanations.

Methods

To answer this research question, the methods SHAP (lecture-covered) and iBreakDown (novel) will be used on the *Network Intrusion Detection* dataset (Abaojiang, 2021). This dataset includes simulated data of a military local network, on which multiple attacks were made. The dataset contains features related to network activity such as duration, protocol and byte sizes of a connection. The data has already been separated into a train and a test set.

Preprocessing & Random Forest

The network dataset includes 42 data columns, which consist of multiple datatypes with most of them being integers and floating-point numbers. Additionally, there are four object-type variables which need to be converted into suitable types that the model can process. The objects can easily be encoded into an interpretable label using the Sklearn LabelEncoder. This results in every changed label being given a value of zero to the number of different labels.

The test set provided does not include the class labels, which means it cannot be used for accuracy measurements. For this reason, the training data has been split into a 20% validation set and an 80% training set. A Random Forest classifier has been fitted on the train set with 100 trees and a maximum tree depth of 10.

Random Forest (RF) has been chosen to use XAI explanations on because of its simplicity and it being a black-box model. A single decision tree can easily be interpreted. However, it suffers from major drawbacks. A single tree is usually inaccurate in class prediction, and it can easily overfit the on the data. The RF model solves this problem by using 100 different trees on varying subsets of the training data. As mentioned, the RF model is a black box, with 100 trees and with 10 levels each there are thousands of decision branches, which means the reason of a classification cannot be traced.

SHAP

The first method of two methods used in this project is SHapley Additive exPlanations or SHAP. As explained in the lecture (*4-Posthoc - 2526 Explainable AI (PER3 V)*, n.d.), SHAP is a model-agnostic, post-hoc explanation based on game theory. It provides local and global explanations of specific observations and the entire model, respectively.

SHAP belongs to the Additive Feature Attribution methods class, this means that the model is based on averaging across, in this case permutations of, features. In contrast, sequential conditioning looks at a specific path of features. It adds features one-by-one to build up its explanation. Being path-dependent means that the explanation changes on what the model has gathered so far about its features.

iBreakDown

The chosen novel method in this project is the iBreakDown algorithm provided by the dalex module. Both SHAP and iBreakDown provide local explanations to observations, they differ however in the mathematical approach used. iBreakDown used sequential conditioning to determine feature importance.

This algorithm works by sequentially adding features one-by-one. It begins with the average prediction in the dataset, the addition each features gives is given by the change each feature

introduces in the model's expected outcome. A significant advantage of using iBreakDown is its ability to detect non-additive interactions. The algorithm can identify interaction effects when the combination of two features significantly differentiates from their individual contribution.

Model Comparison

Feature	SHAP	iBreakDown
Foundation	Game Theory	Sequential conditioning
Strength	Mathematically "fair"	Highly intuitive
Weakness	Computationally expensive	Vulnerable to ordering

Evaluating Methods

Model Fidelity

Both XAI methods provide the impact amount of each feature and its strength on the model's output. To test the model's fidelity, how well the model approximates the black-box predictions, the accuracy is measured of the model without its top three features. This shows the faithfulness of the XAI explanation to the model's internal working.

Cross-Model Consistency

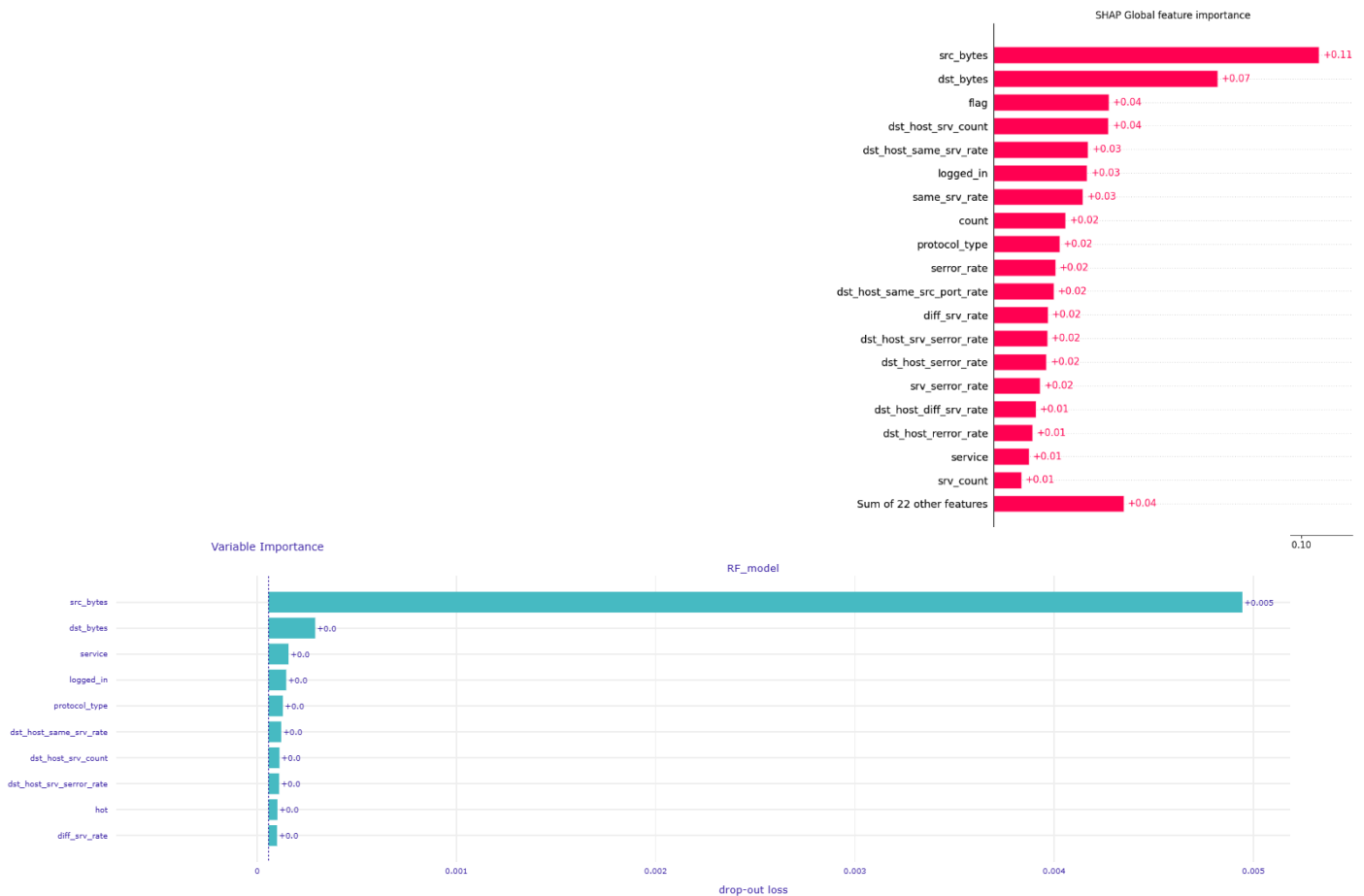
The XAI consistency is checked by keeping track of the agreement rate of both explanations. This rate provides insight into the features that SHAP and iBreakDown use to make their explanations. A high level of agreeance shows the model is robust, with low levels showing the model is too complex to consistently attribute top features.

Results

The random forest model achieved an accuracy of 99.7%, which implies the model is reliable. The following sections give more insight into how this result is made.

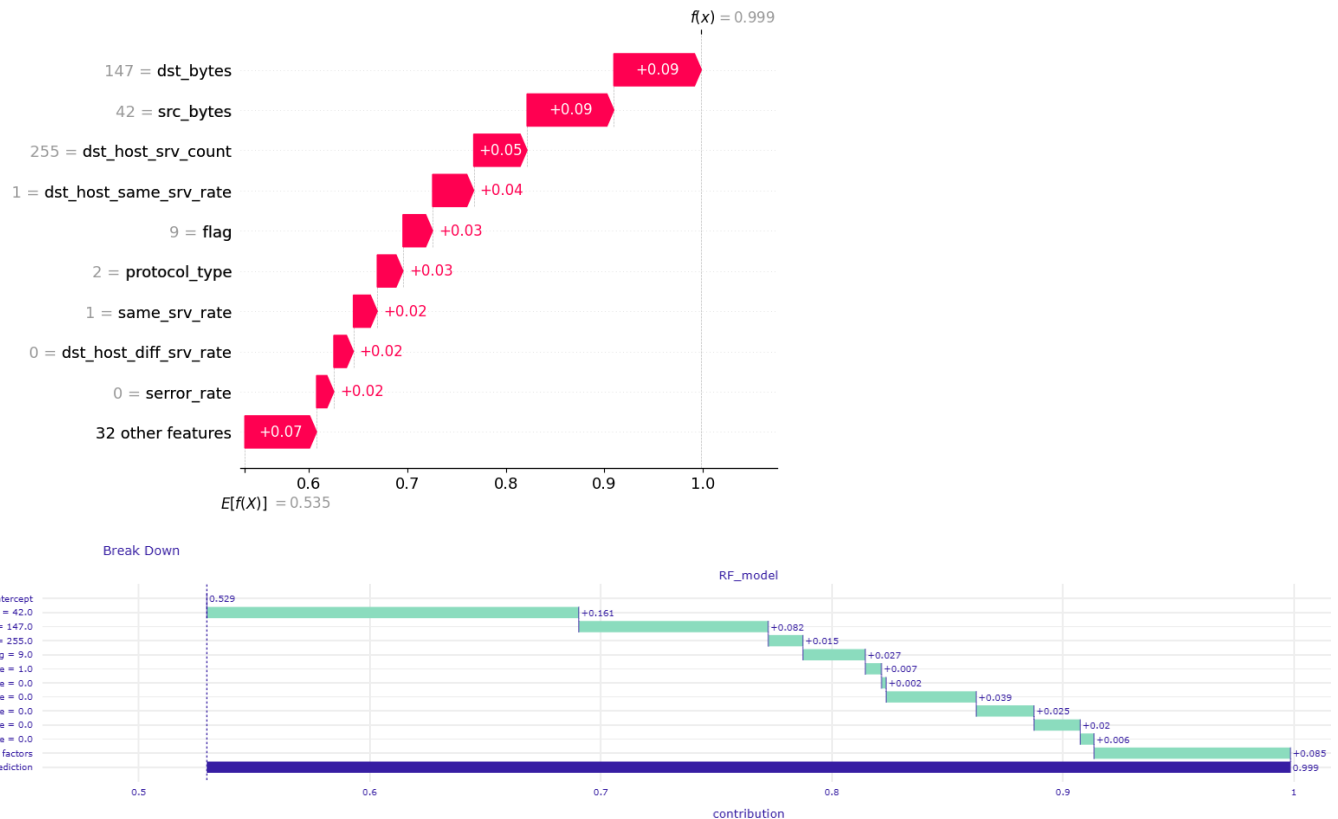
Global explanations

Both algorithms agreed on the most influential feature being *src_bytes* followed by *dst_bytes*. The models do however disagree on the third most important factor. SHAP finds *flag* more important followed closely by *dst_host_srv_count*. While iBreakDown finds all other factors significantly less important than the first two features.



Single instance explanations

To show how each model evaluates a single observation from the data set a waterfall plot is used for SHAP, and the observation is passed as data to the dalex wrapper. One instance where the methods differ is row 355, where SHAP finds *dst_bytes* and *src_bytes* around equally important while preferring *dst_bytes*. iBreakDown significantly prefers *src_bytes* for this instance.



Top feature

The top feature test, where the top three features were masked, shows that the accuracy of the model drops by 10,4%. The top three features are *src_bytes*, *dst_bytes* and *dst_host_srv_count*, and if those features are removed the accuracy drops from 0.99702 to 0.89204, which is a difference of 0.10498 or 10,4%.

Discussion

The models' 99.7% accuracy is supported by the high fidelity. Dropping the top three features' results in a fall in accuracy of 10.4%, this suggests the Random Forest model uses different variables for its predictions showing robustness.

The network intrusion dataset provides some major limitations, such as the collinearity of features. Some features in the dataset measure the same attribute of a cyber-attack, which means part of the variance could be hidden behind similar variables.

Additionally, cyber-attacks usually happen over a longer span of time. The model can only classify packets independently which removes the time-based factor.

References

4-Posthoc - 2526 Explainable AI (PER3 V). (n.d.).

<https://brightspace.ru.nl/d2l/le/content/566875/viewContent/3400370/View>

Abaojiang. (2021, June 14). *Network Intrusion Detection - a simple ML workflow*.

<https://www.kaggle.com/datasets/sampadab17/network-intrusion-detection>

Almheiri, S. J., Shah, A. A., Abbas, S., Ahmad, M., & Khan, M. A. (2025). Smart sustainable cyber security: modelling an interpretable and transparent threat detection with explainable artificial intelligence. *Discover Sustainability*, 6(1). <https://doi.org/10.1007/s43621-025-01280-z>

Gómez, Á. L. P., Maimó, L. F., Celdrán, A. H., & Clemente, F. J. G. (2023). An interpretable semi-supervised system for detecting cyberattacks using anomaly detection in industrial scenarios. *IET Information Security*, 17(4), 553–566. <https://doi.org/10.1049/ise2.12115>

James, C. (2024). Explainable AI for anomaly detection in cybersecurity. *ResearchGate*.

https://www.researchgate.net/publication/387183618_Explainable_AI_for_Anomaly_Detection_in_Cybersecurity

skywqlkered/SOW-BKI266-Explainable-AI: Explaining AI. (n.d.). GitHub.

<https://github.com/skywqlkered/SOW-BKI266-Explainable-AI>